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1. A relational operator
 - a. assigns one operand to another.
 - b. yields a Boolean result.
 - c. compares two operands.
 - d. logically combines two operands.

Ans: b. yields a Boolean result.

2. Write an expression that uses a relational operator to return true if the variable george is not equal to sally.

Ans: George != sally

3. Is -1 true or false?

Ans: True

4. Name and describe the usual purpose of three expressions in a for statement.

Ans: Initialization expression – initializes the loop variable.

Condition (Test) expression – check whether the loop should continue.

Increment/Decrement expression – Updates the loop variable after each iteration

5. In a for loop with a multistatement loop body, semicolons should appear following
 - a. the for statement itself.
 - b. the closing brace in a multistatement loop body.
 - c. each statement within the loop body.
 - d. the test expression.

Ans: c. each statement within the loop body.

6. True or false: The increment expression in a for loop can decrement the loop variable.

Ans: True

7. Write a for loop that displays the numbers from 100 to 110.

Ans: `for(int I = 100; i<= 110; i++)`
`{`
`Cout << I << endl;`
`}`

8. A block of code is delimited by **Curly braces {}**.

9. A variable defined within a block is visible
- a. from the point of definition onward in the program.
 - b. from the point of definition onward in the function.
 - c. from the point of definition onward in the block.
 - d. throughout the function.

Ans: c. from the point of definition onward in the block.

10. Write a while loop that displays the numbers from 100 to 110.

Ans:

```
int I = 100;
while(I <= 110)
{
    Cout << I << endl;
    I++;
}
```

11. True or false: Relational operators have a higher precedence than arithmetic operators.

Ans: False

12. How many times is the loop body executed in a do loop?

Ans: False

13. Write a do loop that displays the numbers from 100 to 110.

Ans:

```
int I = 100;
do
{
    Cout << I << endl;
    I++;
} while(I <= 110);
```

14. Write an if statement that prints Yes if a variable age is greater than 21.

Ans:

```
if(age > 21)
{
    Cout << "yes";
}
```

15. The library function exit() causes an exit from

- a. the loop in which it occurs.

- b. the block in which it occurs.
- c. the function in which it occurs.
- d. the program in which it occurs.

Ans: d. the program in which it occurs.

16. Write an if...else statement that displays Yes if a variable age is greater than 21, and displays No otherwise.

Ans: if (age > 21)

```
{  
    Cout  << "yes";  
}  
Else  
{  
    Cout  << "Nor";  
}
```

17. The getch() library function

- a. returns a character when any key is pressed.
- b. returns a character when Enter is pressed.
- c. displays a character on the screen when any key is pressed.
- d. does not display a character on the screen.

Ans: c. displays a character on the screen when any key is pressed.

18. What is the character obtained from cin when the user presses the Enter key?

Ans: Newline character ('\n')

19. An else always matches the nearest if, unless the if is enclosed in braces.

20. The else...if construction is obtained from a nested if...else by Removing the braces and writing else..if together.

21. Write a switch statement that prints Yes if a variable ch is 'y', prints No if ch is 'n', and prints Unknown response otherwise.

Ans: switch(ch)

```
{  
    case 'y' :  
        cout  << "yes";  
        break;  
    case 'n' :
```

```

        cout << "No";
        break;
    default:
        cout << "unknown response";
    }

```

22. Write a statement that uses a conditional operator to set ticket to 1 if speed is greater than 55, and to 0 otherwise.

Ans: ticket = (speed > 55) ? 1 : 0;

23. The && and || operators

- a. compare two numeric values.
- b. combine two numeric values.
- c. compare two Boolean values.
- d. combine two Boolean values.

Ans: d.combine two Boolean values.

24. Write an expression involving a logical operator that is true if limit is 55 and speed is greater than 55.

Ans: (limit == 55) && (speed > 55)

25. Arrange in order of precedence (highest first) the following kinds of operators: logical, unary, arithmetic, assignment, relational, conditional.

**Ans: 1. Unary
2. Arithmetic
3. Relational
4. Logical
5. Conditional (?:)
6. Assignment (=)**

26. The break statement causes an exit

- a. only from the innermost loop.
- b. only from the innermost switch.
- c. from all loops and switches.
- d. from the innermost loop or switch.

Ans: d. from the innermost loop or switch.

27. Executing the continue operator from within a loop causes control to go to **got to iteration of the loop.**

28. The goto statement causes control to go to

a. an operator.

b. a label.

c. a variable.

d. a function.

Ans: b. a label.